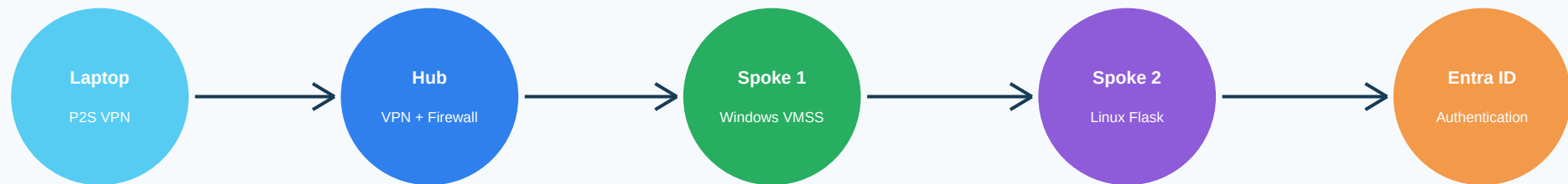


Azure Hub-and-Spoke Project

Visual Quick Guide

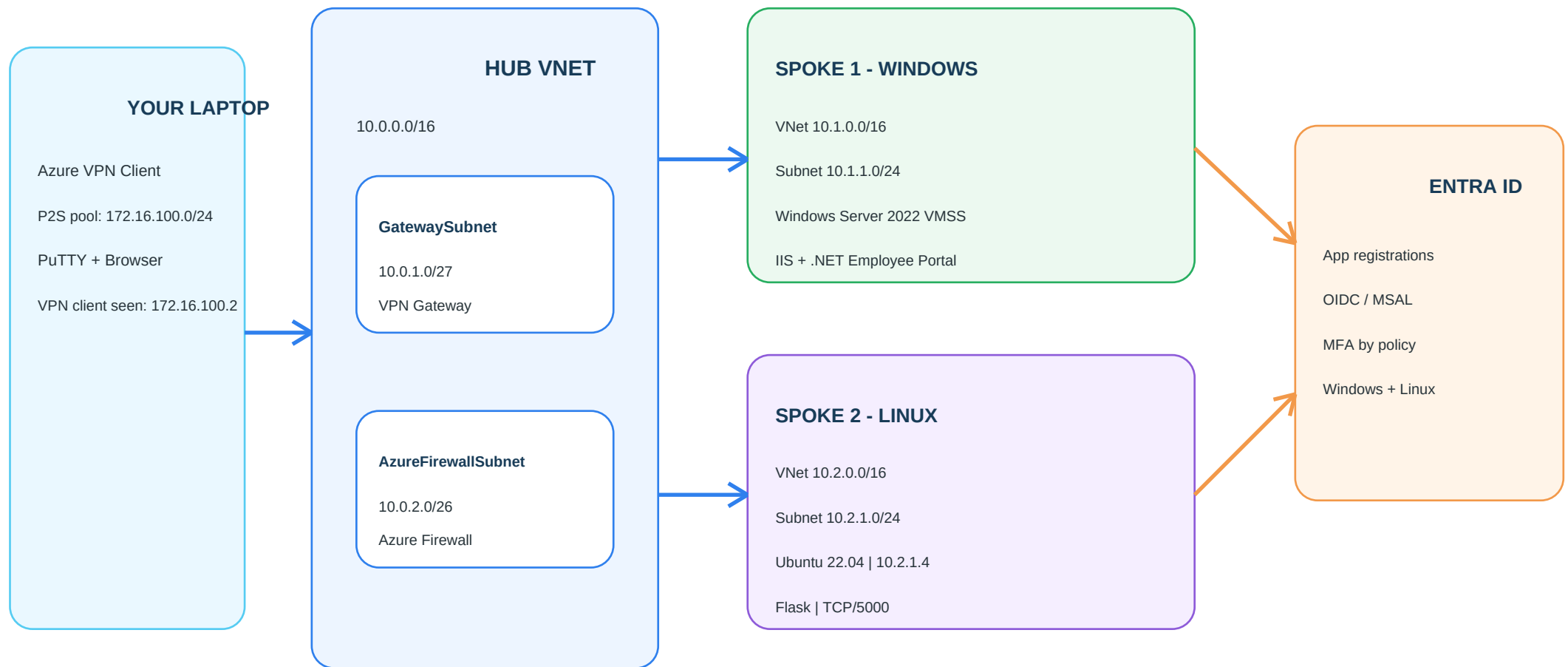
Learn the architecture, deploy the workloads, add Entra ID, test, and clean up.



11 PAGES - LESS TEXT - MORE VISUALS

1. Architecture - Follow the Traffic

Read this page left to right.



Not included: on-premises, DR, Bastion, management subnet, private endpoints.

2. What Each Azure Resource Does

You only need one sentence for each.



VPN Gateway

Private door from your laptop into Azure.



GatewaySubnet

Reserved home for the VPN Gateway.



Azure Firewall

Central traffic-control and inspection point.



VNet Peering

Private bridge between Hub and Spokes.



NSG

Port/source allow-deny rules near a subnet or NIC.



Route Table / UDR

Tells traffic which next hop to use.



VM Scale Set

Multiple similar Windows VMs that can scale.

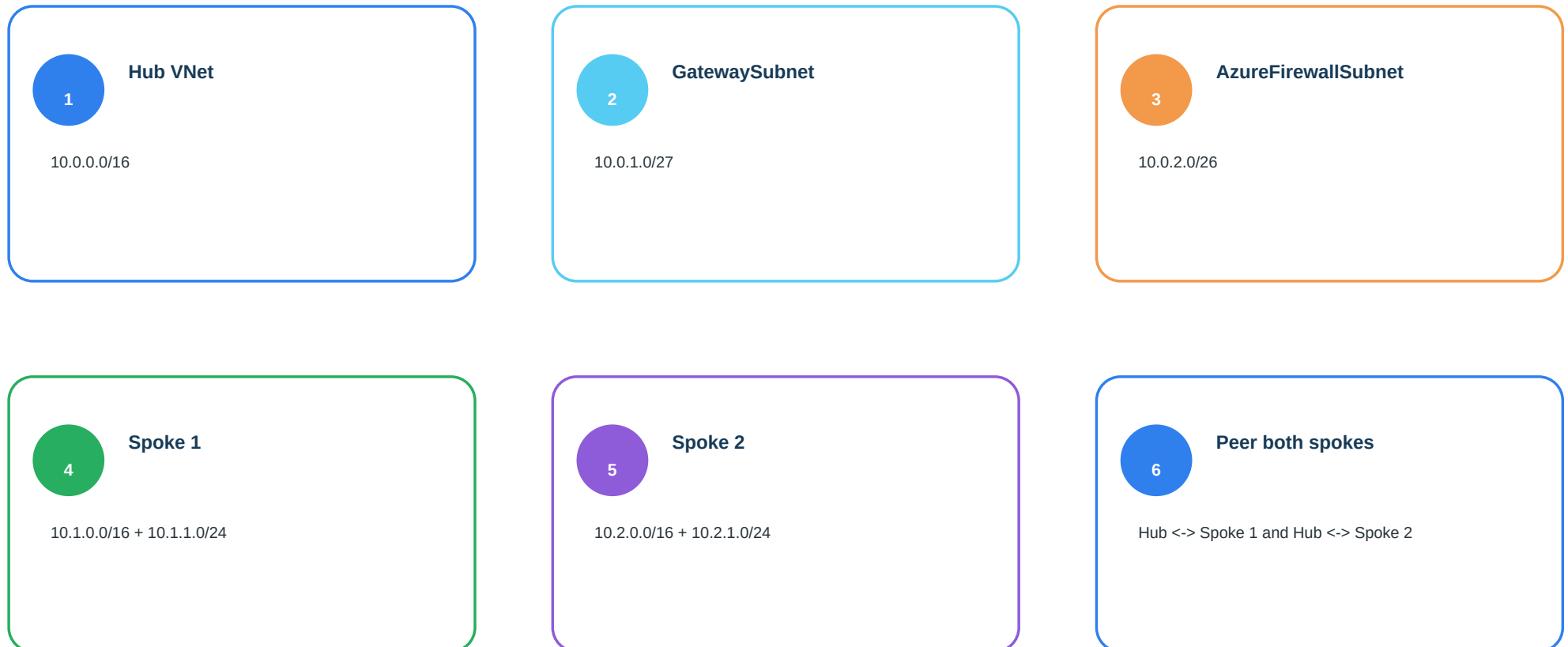


Microsoft Entra ID

Identity provider for app sign-in and MFA policy.

3. Build the Network First

Do this before deploying either application.

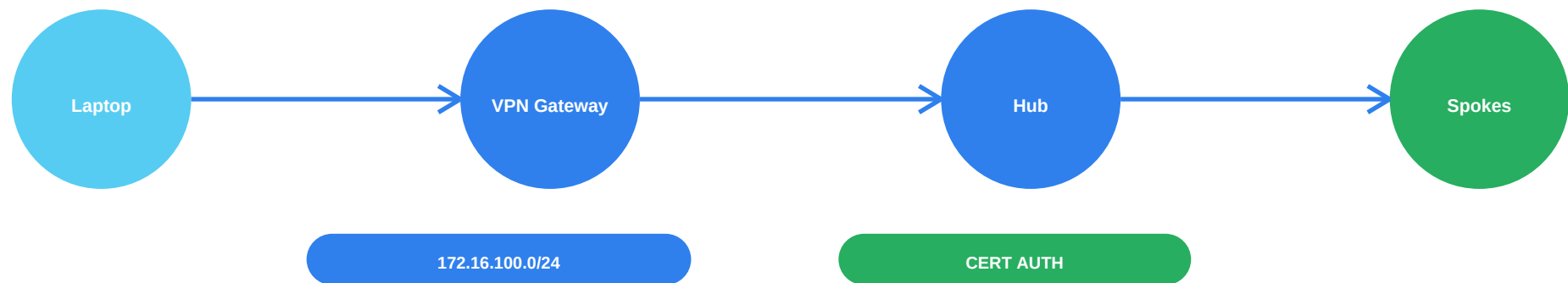


TERRAFORM RHYTHM

```
terraform init      terraform fmt -recursive  terraform validate
terraform plan -out tfplan  terraform apply tfplan
```

4. Connect Your Laptop with P2S VPN

Goal: your laptop gets a private path to both spokes.



WINDOWS TEST

```
Test-NetConnection 10.2.1.4 -Port 22
Test-NetConnection 10.2.1.4 -Port 5000
```

SUCCESS LOOKS LIKE

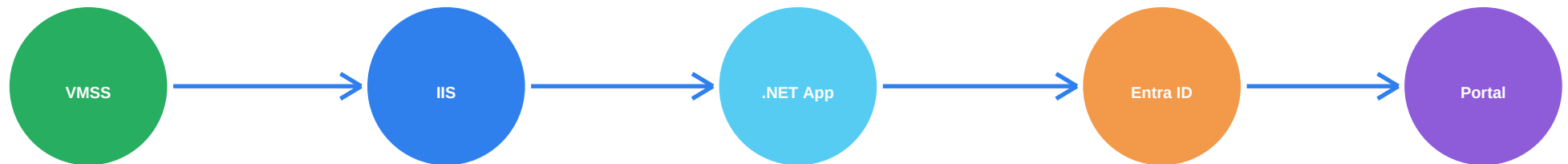
InterfaceAlias: Azure-Hub-Spoke-P2S

SourceAddress: 172.16.100.2

TcpTestSucceeded: True

5. Spoke 1 - Windows VMSS + IIS + .NET

The Windows application path in one picture.



BUILD / PUBLISH

```
cd C:\WebApps\EmployeePortal
dotnet restore
dotnet build
dotnet publish -c Release -o C:\WebApps\EmployeePortal-Publish
```

IIS DEPLOY

```
Import-Module WebAdministration
Stop-WebAppPool -Name "EmployeePortal"
Copy-Item <publish> C:\inetpub\EmployeePortal\ -Recurse -Force
Start-WebAppPool -Name "EmployeePortal"
iisreset
```

Key lesson: Stop the IIS app pool before replacing DLLs; otherwise the files may be locked.

6. Windows App - Entra ID Sign-In

Authentication is a redirect loop, not magic.



APPSETTINGS.JSON

```
"Instance": "https://login.microsoftonline.com/",  
"TenantId": "<TENANT-ID>",  
"ClientId": "<CLIENT-ID>",  
"CallbackPath": "/signin-oidc"
```

COMMON ERROR: AADSTS50011

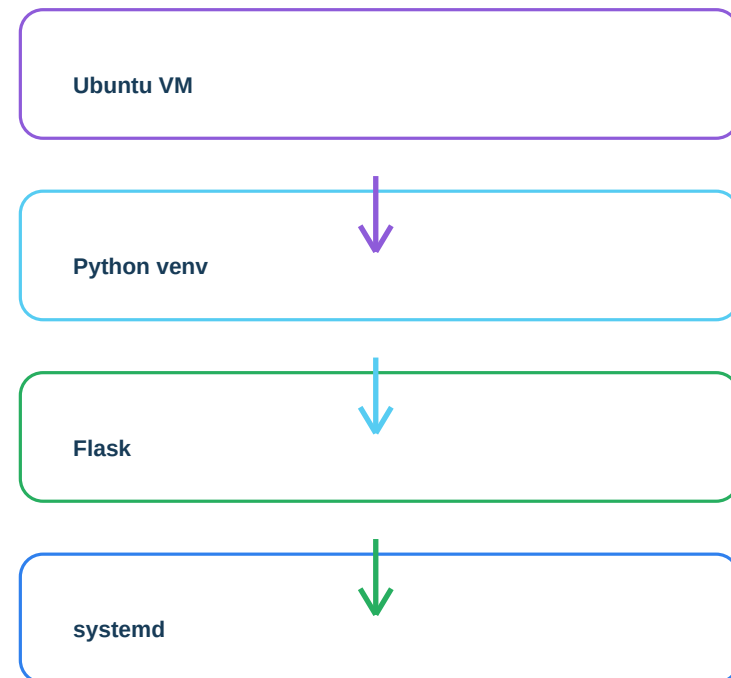
Redirect URI mismatch.

Match scheme + host + port + path exactly.

7. Spoke 2 - Ubuntu + Flask

Private IP: 10.2.1.4 | App port: 5000

```
azureadmin@linuxapp01:~$  
  
hostname  
linuxapp01  
cat /etc/os-release  
PRETTY_NAME="Ubuntu 22.04.5 LTS"  
python3 --version  
Python 3.10.12  
sudo ss -tulpn  
0.0.0.0:22    sshd  
0.0.0.0:5000  python3  
curl http://localhost:5000/health  
{"status": "healthy", "spoke": "spoke-2"}
```



systemd keeps the app running after PuTTY/SSH closes.

8. Linux App - Entra ID with MSAL

Same identity provider, different application stack.



.ENV

```
TENANT_ID=<YOUR-TENANT-ID>
CLIENT_ID=<YOUR-CLIENT-ID>
CLIENT_SECRET=<ROTATED-SECRET>
REDIRECT_PATH=/getAToken
```

PYTHON / SYSTEMD

```
pip install Flask msal python-dotenv requests
sudo systemctl daemon-reload
sudo systemctl restart flaskapp
sudo systemctl status flaskapp --no-pager
```

Security: Never publish a client secret. Revoke any secret that was exposed and create a new one.

9. Access + Troubleshooting

Use the symptom to choose the layer to check.



SSH publickey

Add a new public key; use matching .ppk.

venv failed

Install python3.10-venv and recreate .venv.

App stops

Run it with systemd.

MSAL missing

systemd must use .venv/bin/python.

AADSTS50011

Fix redirect URI exact match.

DLL locked

Stop IIS app pool before copying.

10. Final Checklist + Stop the Cost

If all green, the lab is complete.

- ✓ VPN connected and private routes work
- ✓ Windows VMSS app opens through IIS
- ✓ Windows Entra sign-in works
- ✓ Linux SSH works over VPN
- ✓ Flask /health returns healthy
- ✓ Linux Entra sign-in works
- ✓ Secrets are not stored in the PDF/repository

Project complete when you can explain:

Network -> VPN -> workload -> identity -> test -> cleanup.

WHEN LEARNING IS FINISHED

Always-on services can keep charging.

Keep:

- Terraform files
- Application source
- This visual guide

Then destroy the lab:

CLEANUP

```
terraform plan -destroy
terraform destroy
```